a) $X_a(s) = \dfrac{2s^2 + 7s + 4}{s^3 + 5s^2 + 8s + 4}$

$= \dfrac{3}{s+2} + \dfrac{2}{(s+2)^2} + \dfrac{-1}{s+1}$

inverse laplace transform

$= X_a(t) = 3e^{-2t}u(t) + 2e^{-2t}u(t) - e^{-t}u(t)$

b) $X_b(s) = \dfrac{2s^2+5}{s^2+3s+2} = \dfrac{-13}{s+2} + \dfrac{7}{s+1} + 2$

$X_b(t) = -13e^{-2t}u(t) + 7e^{-t}u(t) + 2\delta(t)$

c) $X_c(s) = \dfrac{8s^2+21s+19}{s^3+5s^2+8s+4} = \dfrac{2}{s+2} + \dfrac{-9}{(s+2)^2} + \dfrac{6}{s+1}$

$x_c(t) = 2e^{-2t}u(t) - 9te^{-2t}u(t) + 6e^{-t}$

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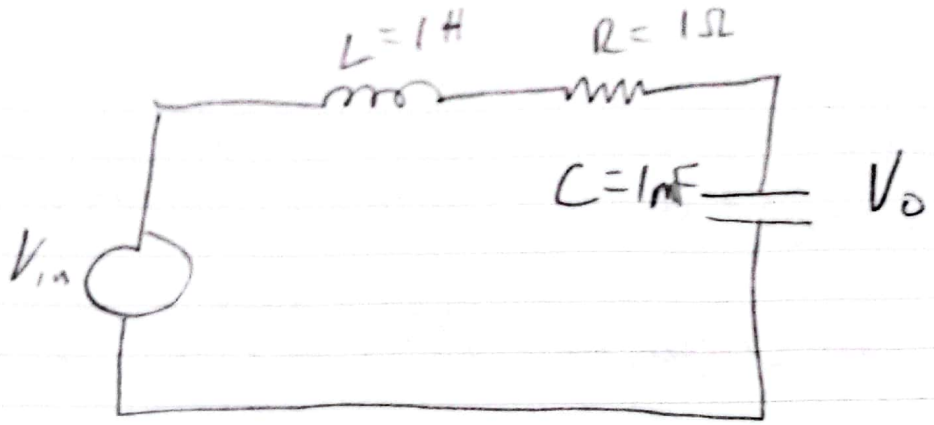
3)



a) $$\boxed{RI + L\frac{dI}{dt} + V_{in} = V_o}$$

b $$V_o = \left(\frac{\frac{1}{j\omega C}}{\frac{1}{j\omega C} + R + j\omega L}\right) V_{in}$$

$$\frac{V_o}{V_{in}} = \frac{\frac{1000}{s}}{\frac{1000}{s} + 1 + s} \quad \frac{s}{s}$$

$$\boxed{\frac{V_o}{V_{in}} = \frac{1000}{s^2 + s + 1000}}$$

$$\frac{W_0^2}{s^2+2\varepsilon W_0 s+W_0^2}$$

$W_0^2 = 1000 \quad W_0 = 31.622$

$$\varepsilon = \frac{1}{2 \cdot 31.622}$$

$= .0158$

$\boxed{\varepsilon = 0.0158}$

system is under damped

c)

$$\frac{1000}{s^2+s+1000}$$

$$\frac{-b \pm \sqrt{b^2-4ac}}{2a}$$

$$s = -\frac{1}{2} + 31.62j \quad , \quad -\frac{1}{2} - 31.62j$$

poles are imaginary and complex

d) For system to be critically damped

$$\varepsilon = \frac{R}{2}\sqrt{\frac{L}{C}} = 1$$

$$\boxed{\begin{aligned} \text{if } R &= 1\,\Omega \\ L &= 8\,H \\ C &= 2\,F \end{aligned}}$$

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